

Course Information

Course Number and Title: **CE 33500 Civil Engineering Materials (Fall 2025)**

CRN information: **27675**

Meeting Day(s) and Time(s):

Lectures

➔ Face-to-Face, Monday, Wednesday, Friday (MWF)

Labs (8 sections)

➔ In-Person, all labs

Note: Before each lab, you are expected to study that materials that will be available in Brightspace. A few i-clicker questions might be asked during the first minutes of your lab session.

Brightspace page: https://purdue.brightspace.com/d2l/home/1357018

Prerequisites: CE 270: Introduction to Structural Mechanics.

Course Description

This course is designed to provide students with basic understanding of relationships between the structure of materials and their properties. It consists of three main parts:

Part I – Fundamentals of Materials Behavior
The course begins with a review of stresses and strains, elasticity, and failure theories in metals from both theoretical and engineering perspectives. The theoretical portion covers atomic bonding, crystalline structures, crystal imperfections and slip systems, and phase diagrams. The engineering aspects are reinforced through laboratory exercises and demonstrations, including tests of compressive, tensile, and torsional behavior; evaluation of elastic modulus; and measurement of hardness and impact resistance.

Part II – Composites, Fracture, and Viscoelasticity
The second part focuses on the behavior of composites, fracture mechanics, and viscoelasticity. Lectures introduce the theoretical background, while laboratories provide hands-on testing of fiber and particulate composites, and demonstrations of fracture and creep behavior.

Part III – Civil Engineering Materials
The final part emphasizes civil engineering materials, including aggregates, portland cement and supplementary cementitious materials (SCMs), concrete, bituminous/asphalt materials and mixtures, and wood. Laboratory work is a central component, with hands-on experiments in aggregate testing as well as the design and testing of asphalt and concrete mixtures.

Course Objectives

By the end of this course, students will be able to apply fundamental concepts of materials science and mechanics to understand, explain, and evaluate the performance of broad range of civil engineering materials, including metals, particulate materials, portland cement and concrete, supplementary cementitious materials (SCMs), asphalt and bituminous concrete, and wood. Students will be able to describe and predict material behavior under arbitrary stress states, and assess failure mechanisms using

mechanics and materials-based arguments supported by microstructural evidence, failure theories, and material composition. In addition, students will be able to design the composition of selected materials, such as portland cement and asphalt concretes, for specific engineering applications.

More specifically, students will be able to:

- Demonstrate a fundamental understanding of the structure-property relationships in various civil engineering materials.
- Describe and predict the mechanical response of materials under different states of stress, including elasticity, yielding, and plasticity.
- Evaluate failure behavior of materials based on mechanics and material science arguments, incorporating microstructural evidence, failure theories, and material composition.
- Successfully implement, analyze, and report on laboratory testing procedures related to civil engineering materials, gaining hands-on experience in material testing techniques.
- Make informed materials selection decisions for civil engineering applications based on material properties and performance characteristics.
- Apply safety protocols and practices when working with materials in laboratory settings, ensuring a safe and productive learning environment.
- Develop critical thinking skills through class activities, quizzes, and discussions, fostering a deeper understanding of course concepts.
- Improve written and verbal communication skills through lab reports, homework assignments, and in-class activities.
- Work effectively in teams with peers on laboratory and coursework tasks while maintaining individual accountability.
- Embrace the core values of civil engineering, including a commitment to ethical practice, societal impact, and sustainability, by integrating these principles into the analysis and selection of civil engineering materials. Understand and apply Purdue CE's distinctive perspectives, such as ***Design Philosophy*** and the ***Civil Engineering Role in Society***, to shape and empower Purdue "Civil" Engineers. This includes recognizing the societal influence and ethical obligations inherent in the field of civil engineering.

The course objectives are designed to guide students in achieving a comprehensive understanding of civil engineering materials, their practical applications, and the essential laboratory skills required to analyze and assess these materials. Students will work towards these objectives through weekly laboratories, in-class activities, lab sessions assignments and laboratory experiments that emphasize hands-on implementation, analysis and reporting of materials testing procedures.

Learning Resources, Technology & Texts

Required texts:

This course does not require a textbook. All course materials and notes will be available on Brightspace: **https://purdue.brightspace.com/d2l/home/1357018**

Additional readings (optional):

- Calister W.D., Rethwish, D.G., “Materials Science and Engineering: An Introduction”
- Mamlouk, M.S., Zaniewski, J. P, “Materials for Civil and Construction Engineers”, 3rd Edition, Pearson, Prentice, Hall ISBM -13 978-0-13-611058-3
- Shackelford, J.F., “Introduction to Materials Science for Engineers”, Prentice Hall, Upper Saddle River, New Jersey.
- "Design and Control of Concrete Mixtures" PCA, Kosmatka and Wilson. (Ed 16th.)
- ACI 211, Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete.

Software/web resources:

You can access the course via Brightspace: https://purdue.brightspace.com/d2l/home/1357018

You will be need Microsoft Word and Microsoft Excel (or similar software) for some lab reports and homework. PDF reader would be also needed.

Learning Outcomes

ABET Student Learning Outcomes Addressed in this Course

#2 An ability to apply engineering *design* to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

#6 An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Course-Specific Learning Outcomes

By the end of the course, you will be able to:

- Formulate equations that describe the properties of composite materials based on boundary conditions, assumptions and constituent properties.
- Design and conduct experiments, analyze and interpret data, and draw meaningful conclusions.
- Learn and apply experimental techniques to characterize material properties and to detect and assess flaws in engineering materials.
- Compare and contrast strength of materials and fracture mechanics solutions.
- Use mathematics, science and engineering principles to estimate the influence of cracks on the mechanical response of engineering materials
- Predict the elastic and viscoelastic responses of materials using mechanics and materials science principles.
- Extract and interpret engineering properties from experimental data.
- Apply standardized test procedures to characterize the properties of engineering materials.
- Compute the influence of material proportions on structure–property relationships in concrete.
- Design concrete mixture proportions for specific engineering applications.
- Evaluate the relationship between asphalt mixture properties and performance.
- Select appropriate engineering materials for specific applications.
- To communicate effectively using graphs.
- Communicate effectively using graphs, executive summaries, and full engineering reports
- Assess and communicate the expected accuracy of results obtained from engineering data.

.GENERAL POLICIES AND RULES

Read these rules CAREFULLY. It will be assumed that you have read this material and understood it.

Attendance

Students are expected to attend all lectures, laboratory sessions, and examinations. Active participation in weekly labs, both problem-solving sessions (HWs) and experimental work, is required. Students should complete assigned readings and watch required videos before coming to class or lab. Unannounced quizzes and in-class activities may occur at any time, making regular attendance essential.

Assignments

Your learning will be assessed through a combination of homework (completed during lab problem -solving sessions), iClicker quizzes, laboratory reports, and two exams (midterm and final) distributed throughout the semester. Detailed information on assignments and exams, including due dates and evaluation rubrics, will be available course website in Brightspace.

The percentage of the total course grade allocated to each type of assignment is shown in the table below.

Assignments	Due	Percent of the grade
Homework (completed and submitted during the lab problem-solving sessions)	Throughout the semester (check homework schedule table on page 12)	20
iClicker quizzes	Throughout the semester	5
Laboratory reports	Throughout the semester	25
Exams	Exam #1 Oct 23rd ,	25
	Exam #2- During the Finals week	25
		Total: 100

Important notes:

- To pass the course, students must earn **at least 100 exam points** out of a total of 200. Failure to meet this minimum will result in a grade of F, regardless of the overall average from other assignments.
- Students are also required to submit **at least 9 out of 11 lab reports** (~82%). Submitting fewer than 82% of the lab reports will result in an automatic grade of F for the course.

Homework - 20 percent of the grade:

(completed and submitted during the lab problem-solving sessions)

To help you stay engaged with the course material and reinforce your learning, we will be replacing homework assignments with **weekly lab problem-solving sessions** this semester. These sessions are designed to give you consistent practice and immediate feedback on key concepts covered in the course

Here's what you need to know:

- **When**: Quizzes will take place during the **first hour of each lab session**.
- **Where**: They must be **completed and submitted in the lab**.
- **Format**:
  - The quizzes are **open notes**, so you can use your written resources to guide your thinking.
  - To support you further, all **necessary equations will be provided** on the quiz sheet.
  - Please remember that **electronic devices** (computers, phones, tablets, etc.) will **not be allowed** during the quiz. This ensures fairness and helps you focus on problem-solving.

Tip for Success: Make a habit of taking thorough notes during lectures, labs, and readings. These notes will be your best resource during quizzes and will help you reinforce the material as you go.

The schedule for the weekly lab problem-solving sessions is included in the schedule document and at the end of the syllabus document. These problem-solving sessions are a valuable opportunity to check your understanding regularly, stay on top of the material, and prepare effectively for exams. Be sure to plan ahead and arrive on time so you can make the most of this learning experience.

Submission guidelines:

- Include your **first and last name, lab section number, and homework number** on each page.
- Bring your **own engineering paper** for completing assignments.
- Begin **each problem on a new page**; staple all pages together before submission.
- Solutions must be **organized**, showing all relevant calculations, notes, sketches, free-body diagrams, Mohr's circles, stresses/strains, etc., as appropriate. Use straight-edges, compasses, and protractors where necessary.
- **Final answers** should be **boxed**, include correct units, and use proper significant figures. Answers without supporting calculations, or with missing/incorrect units, may be graded as incorrect.

Grading:

- Homework will be graded for **correctness or completeness** at the instructor's discretion.
- Solutions will be posted **after assignments are submitted**.

Failure to follow these guidelines may result in **loss of credit**, regardless of answer correctness.

iClicker Quizzes- 5 percent of the grade:

iClicker quizzes will be administered during lectures and lab sessions to actively engage students and reinforce learning. These quizzes provide immediate feedback to both students and the instructor, helping identify areas of misunderstanding and ensuring timely clarification of key concepts. Participation in iClicker quizzes encourages consistent preparation, promotes critical thinking, and enhances comprehension of the material. The quizzes also allow students to gauge their own understanding in real time, supporting more effective learning and retention.

Laboratory reports – 25 percent of the grade

Lab reports are due one week after you perform the experiment or one week after you get all the data needed for the report. Lab reports are due at the beginning of the appropriate lab meeting (by Gradescope). No late reports will be accepted. Please submit a PDF and use the following format for the file:**LastName_FirstInitial_LR#.pdf.** (include your last name, first name initial and replace # with lab session number). All work submitted by the student is expected to be their own effort. If there is reason to believe that work has been copied from another student, university regulations may be invoked as regards punitive action. Furthermore, the Professor reserves the right to assign a failing grade for either the specific work or for the entire course. Problems will be graded for correctness or will be checked for completeness, regardless of approach or solution at the discretion of the Instructor. You need to attend the labs to submit a report. We will not be able to accept any report, if you did not attend your lab session.

Important: Before your in-person Lab Session you must familiarize yourself with the material for that particular lab session and complete the corresponding online quizzes (if any). During the first minutes of the in-person lab sessions, the TA may ask some iClicker questions regarding the lab.

Students are expected, even encouraged, to consult with one another on lab write-ups; however, all work submitted by the student is expected to be their own effort (this includes plots, images, calculations, captions, text, etc.). If there is reason to believe that work has been copied from another student, university regulations may be invoked as regards punitive action. Furthermore, the Professor reserves the right to assign a failing grade for either the specific work or for the entire course.

In addition, students should **consult the section titled "Responsible use of AI in Completing Coursework",** provided later in this syllabus, on responsible use of the Artificial Intelligence (AI) in completing the coursework. Misuse of AI tools in coursework, which includes but is not limited to producing unoriginal work, uncited use of AI-generated content, or unauthorized assistance on assessments, will be considered a breach of academic integrity. Consequences will follow the Purdue's policies on academic dishonesty as detailed in this syllabus, which may include grade penalties, course failure, or more severe disciplinary actions.

Exams (50 points). All exams are closed book, closed notes. Relevant formulas will be provided with the exam. Programmable calculators are permitted; however, grading will be based solely on the information shown on the exam sheet. For this reason, all necessary steps, figures, and calculations are to be shown in order to obtain credit. The correct answer does not ensure that you will receive full credit – **You must provide sufficient detail to show how the solution was obtained.** All work that you submit must be strictly your own. If there is reason to believe that work has been copied or done in collaboration with another student, university regulations may be invoked as regards punitive action. **Instances of cheating during exams will result in full loss of credit for that exam.** Additional measures including the immediate failure of the course, may be applied at the discretion of the instructor and/or University Staff.

Make-Up Exams.
There will be NO MAKE UP EXAMS in this course. Any student absent from an exam will receive a score of "zero" for the exam, unless the absence was related to a substantive personal or family health emergency. The dates of the exams are indicated on the syllabus. Therefore, all travel plans associated within or outside the university functions shall be made with this policy in mind (this includes the second exam offered during the Finals week). Any student missing an exam due to a substantive personal or family health emergency shall provide a written letter from a physician or a recognized individual authenticating the seriousness of the emergency and how it led to the student's absence from the exam. The letter should state the specific reason for the absence and the date and duration of the incident. The letter shall include the name, title, relationship to the student, address, and telephone number of the letters author. In such case, the instructor reserves the right to accept or decline the excuse.

Grading Scale

In order to pass the course, a letter grade of C is required for the final course grade. The course is NOT graded on a CURVE. Final letter grades are assigned based on the total number of percentage points accumulated. These percentage points are weighted so that the approximate breakdown looks as follows: As a guide, letter grades will generally be assigned as follows: **Total percentage in the 90's- A, 80's- B, 70's- C, 60's- D, below 60's- F;**

You are welcome to discuss any grade with Dr. Olek. **It is important that this is done within 72 hours of when the graded work is returned**. There will be no general review of prior work to find "extra points" toward the end of the semester. You are free to contest laboratory, homework, or exam grades; however, this must be done in writing and submitted to Dr. Olek with a copy to your assigned TAs. This written explanation should include the location of the problem you are requesting review for, along with the information as to where the error is believed to have occurred.

Good grades are earned by students who (i) attend and participate in course activities, (ii) pay attention during the lectures, (iii) perform the assigned homework and lab reports, (iv) understand the fundamental concepts and objectives presented in class and demonstrate it through homework, lab reports, quizzes and exams.

It is important that you always check your grades on Brightspace, and estimate your own overall grade. If you have questions or concerns about your grades, or performance in general, contact Prof. Olek.

Responsible use of AI in Completing Coursework

Use of AI tools for some assignments and lab reports is not banned but they must be used responsibly and ethically. Much of the work must be the student's own. For grading consistency, only ChatGPT is permitted.

- All use of AI must be fully disclosed in submissions. At a minimum, this includes providing the prompts used, and when appropriate, the chatbot responses as well.
- Failure to disclose AI use will result in penalties ranging from a zero on the assignment to a reduction in the final course grade. Repeated violations may result in failure of the course, regardless of total scores.
- Even with full disclosure, excessive reliance on AI (e.g., copying extensive chatbot responses directly) will be penalized, though less severely than undisclosed use.
- Students are encouraged to consult the instructor or teaching assistants if clarification is needed.

Academic Integrity

Academic integrity is expected of all students at all times. Information on what constitutes academic integrity can be found on the web site at the following url:
https://www.purdue.edu/odos/osrr/resources/documents/academic_integrity.html

Additional information is provided in the "University Regulations" located at:
https://catalog.purdue.edu/content.php?catoid=13&navoid=15965

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace table of contents, under University Policies.

Students found guilty of academic dishonesty will be reported to the Office of the Dean of Students and will automatically receive a failing grade from the course.

Purdue's Honor Pledge: "As a boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue."

Use of Copyrighted Materials

Among the materials that may be protected by copyright law are the lectures, notes, videos, assignments, exams, and other material presented in class or as part of the course. Always assume the materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor. To obtain permission to sell or barter notes, the individual wishing to sell or barter the notes must be registered in the course or must be an approved visitor to the class. Course instructors may choose to grant or not grant such permission at their own discretion, and may require a review of the notes prior to their being sold or bartered. If they do grant such permission, they may revoke it at any time, if they so choose.

Violent Behavior Policy

Purdue University is committed to providing a safe and secure campus environment for members of the university community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent Behavior impedes such goals. Therefore, Violent Behavior is prohibited in or on any University Facility or while participating in any university activity. See the following website for additional information: https://www.purdue.edu/policies/facilities-safety/iva3.html.

Accessibility

Purdue University strives to make learning experiences accessible to all participants. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at **drc@purdue.edu** or by phone: 765-494-1247, as soon as possible.
If the Disability Resource Center (DRC) has determined reasonable accommodations that you would like to utilize in my class, you must release your Course Accommodation Letter to me. Instructions on sharing your Course Accommodation Letter can be found by visiting: **https://www.purdue.edu/drc/students/course-accommodation-letter.php** Additionally, you **are strongly encouraged to contact me as soon as possible** to discuss implementation of your accommodations

Nondiscrimination Statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Purdue University prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. The University will conduct its programs, services and activities consistent with applicable federal, state and local laws, regulations and orders and in conformance with the procedures and limitations as set forth in Executive Memorandum No. D-1, which provides specific contractual rights and remedies. Any student who believes they have been discriminated against may visit www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

Academic Guidance in Event of Qualified Illness

If you find yourself too sick to progress in the course, notify your advisor and notify me via email or Brightspace. We will make arrangements based on your particular situation.

Mental Health

If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a new web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS)**.** TAO is available to students, faculty, and staff at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Basic Needs Security:

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday.

Emergency Procedures

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances. Information will be provided via email and/or Brightspace. Please see Purdue's Emergency Preparedness website at https://www.purdue.edu/ehps/emergency_preparedness/.

Emergency exist plans have been prepared for all classrooms located in engineering buildings. The following link provides critical information Evacuation and Shelter-in-place for the RHPH--ROBERT HEINE PHARMACY BUILDING
https://www.purdue.edu/ehps/emergency-preparedness/emergency-plans/bep/building-beps/rhph-bep.html

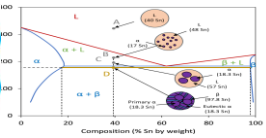
Course Schedule

Purdue Academic Calendar link and key University dates for the Fall 2025semester are included here:

- August 28, 2025 – Classes Begin
- September 1, 2025 – LABOR DAY (No classes)
- October 13, 14, 2025 – OCTOBER BREAK (No classes)
- October 23, 2025 – 1st Exam (PHYS 114, 8:00 pm -10:00 pm)
- November 26-29, 2025, - THANKSGIVING BREAK (No classes)
- December 13, 2025 – Classes End
- December 15-20, 2025– 2nd Exam (Time and location TBD)

Others

No food in the lab. No photos or filming with your phones or other devices in the classroom or lab.






Schedule: Lectures and Labs CE33500- CRN 27675 (Fall 2025)

Week #	Date	Day	Lecture	Lecture Title
Week 1 (Aug 25 - Aug 29)	08/25	M	L1	Introduction & Course Syllabus
	08/27	W	L2	Review of concepts: Force, stress, strain
	08/29	F	L3	Review of concepts: Principal stresses and Mohr's circle
	~		~	**NO LAB**
Week 2 (Sep 1 – Sep 5)	09/01	M	-	NO CLASS - LABOR DAY
	09/03	W	L4	Elasticity & Hooke's Law (I)
	09/05	F	L5	Elasticity and Hooke's Law (II)
	~	**Check group**	**L1, H1**	**Machine operation & Materials variability-Formal Lab Report**
Week 3 (Sep 8 – Sep 12)	09/08	M	L6	Plasticity, True stress/strain Generalized Hooke's Law
	09/10	W	L7	Theories of Failure
	09/12	F	L8	Atomic bonding (I)
	~	**Check group**	**L2, H2**	**Testing to determine stress, strain and E-modulus**
Week 4 (Sept 15 – Sep 19)	09/15	M	L9	Atomic Bonding (II), Crystal Structure I
	09/17	W	L10	Crystal Structure II
	09/19	F	L11	Crystal Structure III (dir. & planes)
	~	**Check group**	**L3, H3**	**Torsion**
Week 5 (Sept 22 – Sep 26)	09/22	M	L12	Imperfections in Crystals (I)
	09/24	W	L13	Imperfections in Crystals (II);Slip systems in Crystals (I)
	09/26	F	L14	Slip systems in Crystals (II); Phase diagrams I
	~	**Check group**	**L4, H4**	**Hardness and Impact Testing**
Week 6 (Sep 29 – Oct 3)	09/29	M	L15	Phase Diagrams II
	10/01	W	L16	Phase Diagrams III
	10/03	F	L17	Composites I
	~	**Check group**	**L5**	**Tensile testing of steel, PVC and acrylic samples**
Week 7 (Oct 6 – Oct 10)	10/06	M	L18	Composites II
	10/08	W	L19	Viscosity &Viscoelasticity I
	10/10	F	L20	Viscoelasticity II
	~	**Check group**	**L6, H5**	**Fiber and particulate composites**
Week 8 (Oct 13 – Oct 17)	10/13	M	-	NO CLASS – OCTOBER BREAK
	10/15	W	L21	Viscoelasticity III
	10/17	F	L22	Fracture I
	~	**Check group**	**L7, H6**	**Viscoelasticity (creep) – Virtual Lab. – Formal Lab Report**
Week 9 (Oct 20– Oct 24)	10/20	M	L23	Fracture II
	10/22	W	L24	Fracture III
	10/23	TH	~	**FIRST EXAM -**
	10/24	F	L25	Fracture IV and Aggregate I
	~	**Check group**	**L8, H7**	**Fracture toughness testing**
Week 10 (Oct 27 – Oct 31)	10/27	M	L26	Aggregate II
	10/29	W	L27	Aggregate III
	10/31	F	L28 & L29	Asphalt I (Bituminous Materials) & Asphalt II (Mixtures)
	~	**Check group**	**L9, H8**	**Aggregate gradation and density**
Week 11 (Nov 3 – Nov 7)	11/03	M	L30 & L31	Asphalt III (Mixture Design) & Asphalt IV (Plant Operation)
	11/05	W	L32	Summary and Comments on Asphalt Lectures
	11/07	F	L33	Concrete I -Introduction-
	~	**Check group**	**L10, H9**	**Asphalt Samples Preparation**

Week 12 (Nov 10 – Nov 14)	11/10	M	L34	Concrete II & Cement I
	11/12	W	L35	Concrete Mixture Proportioning
	11/14	F	L36	Concrete Mixture Proportioning – In class exercise
	~	**Check group**	**L11, H10**	**Asphalt Sample testing - Formal Lab Report (L9&10)**
Week 13 (Nov 17 – Nov 21)	11/17	M	L37	Cement II
	11/19	W	L38	Cement III & SCMs I
	11/21	F	L39	SCMs II
	~	**Check group**	**L12, H11**	**Concrete Samples Preparation**
Week 14 (Nov 24 – Nov 28)	11/24	M	~	NO CLASS – OMITTED TO OFFSET THE EVENING EXAM
	11/26	W	~	NO CLASS – THANKSGIVING VACATION
	11/28	F	~	NO CLASS – THANKSGIVING VACATION
	~		**NO LAB**	**-THANKSGIVING VACATION**
Week 15 (Dec 1 – Dec 5)	12/01	M	L40	Concrete – Early Age Properties I
	12/03	W	L41	Concrete-Early Age Properties (II) & Mechanical Properties I
	12/05	F	L42	Concrete- Mechanical Properties II
	~	**Check group**	**L13, H12**	**Concrete Samples Testing - Formal Lab Report (L12&13)**
Week 16 (Dec 8 – Dec 12)	12/08	M	L43	Concrete – Mechanical Properties II & Durability I
	12/10	W	L44	Concrete – Durability II & Wood and Timber I
	12/12	F	L45	Wood & Timber II & Masonry
	~		**NO LAB**	**-**
SECOND EXAM- During the Finals Week (Dec 15-20)				

Tentative Schedule of Homework (Lab Problem-Solving Sessions)

HW No.	Homework title	Dates
1	Stress transformation equations	**Week 2** (Sep 1 – Sep 5)
2	Principal stresses and Mohr's circle	**Week 3** (Sep 8 – Sep 12)
3	True stress/strains and theories of failure	**Week 4** (Sept 15 – Sep 19)
4	Atomic bonding, crystal structure and Miller Indices	**Week 5** (Sept 22 – Sep 26)
5	Phase diagrams & crystal imperfections	**Week 7** (Oct 6 – Oct 10
6	Composites (**virtual**)	**Week 8** (Oct 13 – Oct 17)
7	Viscoelasticity	**Week 9** (Oct 20– Oct 24)
8	Fracture mechanics	**Week 10** (Oct 27 – Oct 31))
9	Aggregates	**Week 11** (Nov 3 – Nov 7)
10	Asphalt	**Week 12** (Nov 10 – Nov 14)
11	Concrete I- Mixture design	**Week 13** (Nov 17 – Nov 21)
12	Concrete II- hydration and mechanical properties	**Week 15** (Dec 1 – Dec 5)

Disclaimer:

This schedule is subject to change.